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ORIGINAL PAPER

# Thermodynamic description of the Ce–Ga and Ga–Gd systems

**Jinming Liu[1] · Fei Li[1] · Jinping Liu[1] · Xu Zhang[1] · Tahir Ahmad[1] · Tongxiang Liang[1]**

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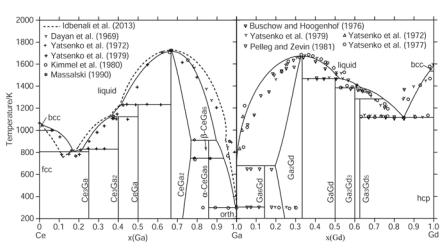
## Abstract

The Ce–Ga and Ga–Gd systems are critically assessed by means of the calculation of phase diagram (CALPHAD) technique. The excess Gibbs energy of the solution phases [liquid, orthorhombic (Ga), fcc (Ce), bcc (Ce or Gd), and hcp (Gd)] is modeled with the Redlich–Kister equation. In the Ce–Ga and Ga–Gd systems, the intermetallic compounds, CeGa, $Ce_3Ga_2$, $Ce_3Ga$, α-$CeGa_6$, β-$CeGa_6$, GaGd, $Ga_6Gd$, $Ga_2Gd_3$, and $Ga_3Gd_5$ are treated as stoichiometric compounds. The intermetallic compounds $CeGa_2$ and $Ga_2Gd$ with a homogeneity range, are treated using a two-sublattice model as the formula $(Ce,Ga)_{0.333333}(Ce,Ga)_{0.666667}$, and $(Ga,Gd)_{0.666667}(Ga,Gd)_{0.333333}$, respectively. In the Ce–Ga system, there are nine invariant reactions, and five stoichiometric intermetallic compounds and the compound $CeGa_2$ with a homogeneity domain ranging from about 66.67 to 77.94 at.% Ga. Some improvements are obtained based on the experimental data in the present work, the liquidus line is more close to the most experimental points, and the homogeneity range of the compound $CeGa_2$ seems more reasonable at full temperature range. In the Ga–Gd system, there are eight invariant reactions, and four stoichiometric intermetallic compounds and the compound $Ga_2Gd$ with a homogeneity domain ranging from about 19.64 to 33.33 at.% Gd. The present phase diagram and thermodynamic properties of the system are well reproduced with the experimental data.

### Graphical abstract






**Keywords** Ce–Ga system · Ga–Gd system · Thermodynamic assessment · Thermodynamic properties · CALPHAD technique

## Introduction

The multi-alloys of gallium with lanthanides are of high probability to use due to the production of good magnetic materials [1, 2]. The magnetocaloric effect (MCE) includes the temperature change of the magnetic material under the action of an external magnetic field. The main role of the MCE is based on promising and environmentally friendly magnetic refrigeration technology [1]. The challenge is to have high-magnetic refrigeration efficiency near room temperature at relatively low applied magnetic fields. While the thermodynamic properties of the Ga-based alloys are

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✉ Jinming Liu
liujm2011@jxust.edu.cn

[1] School of Material Science and Engineering, Jiangxi University of Science and Technology, Ganzhou 341000, China

necessary information for the improvement of the MCE. Therefore, to get the thermodynamic information about the Ga-based alloys, it is necessary to thermodynamically assess the Ga-based systems.

This study is intended to give a better understanding of the thermodynamic properties of the Ga-based alloys. Thermodynamic description is also needed to obtain the thermodynamic factors via the CALPHAD technique. The Ce–Ga system was assessed with an association model of the liquid by Idbenali et al. [3]. But in the assessed Ce–Ga system, the compound $CeGa_2$ was almost no solid solubility at high temperature [3], which is greatly different from the phase diagram evaluated by Massalski [4], as shown in the reproduced results using the thermodynamic parameters by Idbenali et al. [3] in Fig. 1. So the Ce–Ga system is re-assessed based on the reported experimental information in the present work. No thermodynamic modeling of the Ga–Gd system is available in the literature. In the present work, the liquid and solid solutions of the Ce–Ga and Ga–Gd systems are modeled with the Redlich–Kister model [5].

## Literature review

### The Ce–Ga system

All the information of the Ce–Ga system was compiled in detail by Idbenali et al. [3]. The early Ce–Ga system was determined by Yatsenko et al. [6], including four compounds, $Ce_3Ga$, $Ce_5Ga_3$, CeGa, and $CeGa_2$. In 1979, the

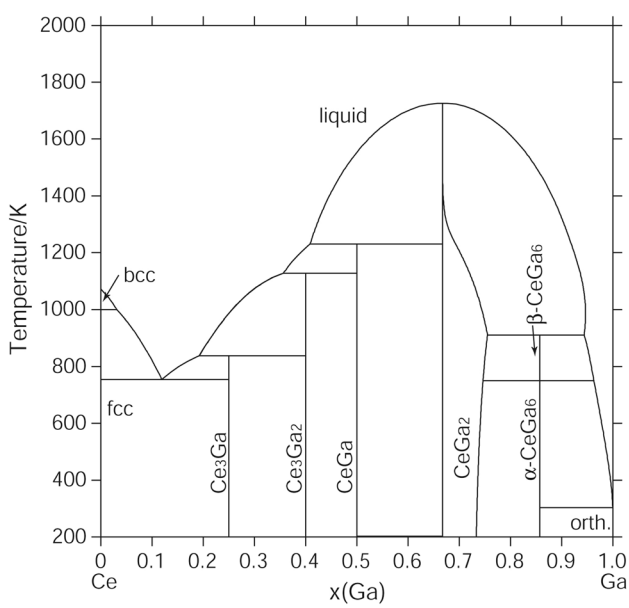


**Fig. 1** Reproduced Ce–Ga phase diagram using the thermodynamic parameters reported by Idbenali et al. [3]

Ce–Ga alloys were investigated, and the compounds $Ce_3Ga_2$ and CeGa (decomposition at 850 and 960 °C, respectively) and $CeGa_2$ (melting point at 1444 °C) were found by Dayan and Pelleg [7]. Later on, the compound $CeGa_6$ with the homogeneity range of $CeGa_2$ from about 66.67 to 78 at.% Ga was found and formed by a peritectic reaction at 453 ± 7 °C by Kimmel and Dayan [8]. The Ce–Ga phase diagram was compiled by Massalski [4] based on the above Refs [6–8]. The liquidus of the Ga-rich side of the Ce–Ga diagram was determined using the solubility for Ce in liquid Ga by Yatsenko et al. [9].

Thermodynamic information of the Ce–Ga system were investigated by several researchers [10–16] and all the thermodynamic information were compiled in detail by Idbenali et al. [3]. The enthalpy of formation of $CeGa_2$ and the cerium partial enthalpy of mixing at infinite dilution were determined using the electro-motive force (EMF) technique in the 675–975 K range by Vnuchkova et al. [10]. The enthalpies of dissolution of cerium in gallium at 1213 K, the enthalpy of formation of $CeGa_2$, and the solubility of cerium in liquid Ga were determined by Yamshchikov et al. [11, 12]. The enthalpies of mixing of Ce–Ga alloys from 0 to 0.12 at.% Ce at 1250 K were determined by Chkhikvadze and Kereselidze [13]. The enthalpies of formation of all the intermetallic compounds in the Ce–Ga system were calculated via a semi-empirical model by Boer et al. [14]. In 2001, the enthalpies of formation were determined using different methods by Babu et al. [15] and Meschel and Kleppa [16]. The standard enthalpies of formation of $Ce_3Ga_2$, CeGa, $CeGa_2$, and $CeGa_6$ were determined by high-temperature liquid gallium solution calorimetry [15, 16]. The standard enthalpy of formation of $Ce_{0.21}Ga_{0.79}$ at 1125 K and $CeGa_6$ at 876 K was determined using precipitation calorimetry [15]. The integral enthalpies of formation of Ce–Ga alloys at 1125 and 876 K for (0.061–24.12 at.% Ce) using precipitation calorimetry and the partial enthalpy of solution of cerium in liquid gallium at infinite dilution at 874 and 1094 K [15].

### The Ga–Gd system

All the experimental data of the Ga–Gd phase diagram, compounds, thermodynamic, and magnetic properties were compiled in detail by Palenzona and Clrafici [17]. The phase relations of the Ga–Gd system were determined experimentally by Yatsenko et al. [6], Buschow [18], Yatsenko [19], and Pelleg et al. [20]. In 1990, an assessed phase diagram was constructed by Palenzona and Clrafici [17] based on the experimental data [6, 18–20]. The crystal structure and lattice parameter data for α-Ga (orth.), α-Gd (hcp_A3), β-Gd (bcc), $Ga_2Gd$, GaGd, $Ga_2Gd_3$, $Ga_6Gd$, and $Ga_3Gd_5$ were confirmed in many investigations [6, 18, 20–29]. The homogeneity range of the compound $Ga_2Gd$ was about 11.3 at.% Gd, extending to the Ga-side between 22.0 and

**Table 1** Thermodynamic parameters of the Ce–Ga and Ga–Gd systems (J mol$^{-1}$)

| Systems | Phases | Model | Thermodynamic parameters |
|---|---|---|---|
| Ce–Ga | Liquid | $(Ce, Ga)_1$ | ${}^0L^{liq} = -253596.1+63.8096T - 0.0013T\ln(T) - 0.0049T^2$ |
| | | | ${}^1L^{liq} = +81990.8 - 0.4089T$ |
| | | | ${}^2L^{liq} = -31446.3$ |
| | | | ${}^3L^{liq} = +31966.8$ |
| | Orth. | $(Ce, Ga)_1$ | G(orth., Ce) = + $GHSER_{Ce}$+5000.0 |
| | | | G(orth., Ga) = + $GHSER_{Ga}$ |
| | bcc | $(Ce, Ga)_1$ | G(bcc, Ce) = + GHBCCCE |
| | | | G(bcc, Ga) = + GHBCCGA |
| | fcc | $(Ce, Ga)_1$ | G(bcc, Ce) = + $GHSER_{Ce}$ |
| | | | G(bcc, Ga) = + $GHSER_{Ga}$+3800.0 − 10.2000$T$ |
| | α-$CeGa_6$ | $(Ce)_{0.142857}$ $(Ga)_{0.857143}$ | $G^{\alpha\text{ - }CeGa_6}_{Ce:Ga} = +0.142857GHSER_{Ce} + 0.857143GHSER_{Ga} - 56904.0+10.3472T$ |
| | β-$CeGa_6$ | $(Ce)_{0.142857}$ $(Ga)_{0.857143}$ | $G^{\beta\text{ - }CeGa_6}_{Ce:Ga} = +0.142857GHSER_{Ce} + 0.857143GHSER_{Ga} - 52176.0+4.0285T$ |
| | $Ce_3Ga$ | $(Ce)_{0.75}$ $(Ga)_{0.25}$ | $G^{Ce_3Ga}_{Ce:Ga} = +0.75GHSER_{Ce} + 0.25GHSER_{Ga} - 50397.0+16.7494T$ |
| | $Ce_3Ga_2$ | $(Ce)_{0.6}$ $(Ga)_{0.4}$ | $G^{Ce_3Ga_2}_{Ce:Ga} = +0.6GHSER_{Ce} + 0.4GHSER_{Ga} - 65050.1+9.4306T$ |
| | CeGa | $(Ce)_{0.5}$ $(Ga)_{0.5}$ | $G^{CeGa}_{Ce:Ga} = +0.5GHSER_{Ce} + 0.5GHSER_{Ga} - 75748.6+10.1322T$ |
| | $CeGa_2$ | $(Ce, Ga)_{0.333333}$ $(Ce, Ga)_{0.666667}$ | $G^{CeGa_2}_{Ce:Ga} = +0.333333GHSER_{Ce} + 0.666667GHSER_{Ga} - 94018.9+13.6124T$ |
| | | | $G^{CeGa_2}_{Ce:Ga} = +0.333333GHSER_{Ga} + 0.666667GHSER_{Ce} + 10233.4$ |
| | | | $G^{CeGa_2}_{Ce:Ce} = +GHSER_{Ce} + 50243.3$ |
| | | | $G^{CeGa_2}_{Ga:Ga} = +GHSER_{Ga} + 10234.3$ |
| | | | ${}^0L^{CeGa_2}_{Ce,Ga:Ga} = -51685.7 - 3.2953T$ |
| | | | ${}^0L^{CeGa_2}_{Ce,Ga:Ce} = +0.0$ |
| | | | ${}^0L^{CeGa_2}_{Ce:Ce,Ga} = {}^0L^{CeGa_2}_{Ga:Ce,Ga} = +0.0$ |
| Ga–Gd | Liquid | $(Ga, Gd)_1$ | ${}^0L^{liq} = -118889.7+2.4170T$ |
| | | | ${}^1L^{liq} = -37609.8+28.9465T$ |
| | | | ${}^2L^{liq} = -2392.0+1.5000T$ |
| | Orth. | $(Ga, Gd)_1$ | G(orth., Gd) = + $GHSER_{Gd}$+5000.0 |
| | | | G(orth., Ga) = + $GHSER_{Ga}$ |
| | bcc | $(Ga, Gd)_1$ | G(bcc, Gd) = + $GHSER_{Gd}$ |
| | | | G(bcc, Ga) = + $GHSER_{Ga}$+4500.0 − 11.7000$T$ |
| | hcp | $(Ga, Gd)_1$ | G(hcp, Gd) = + $GHSER_{Gd}$+3572.1 − 2.30507$T$ |
| | | | G(hcp, Ga) = + $GHSER_{Ga}$+4500 − 9.5000$T$ |
| | $Ga_2Gd$ | $(Ga,Gd)_{0.666667}(Ga,Gd)_{0.333333}$ | $G^{Ga_2Gd}_{Gd:Ga} = +0.666667GHSER_{Gd} + 0.333333GHSER_{Ga}+43043.4$ |
| | | | $G^{Ga_2Gd}_{Ga:Gd} = +0.666667GHSER_{Ga} + 0.333333GHSER_{Gd} - 61746.1+6.5034T$ |
| | | | $G^{Ga_2Gd}_{Ga:Ga} = +GHSER_{Ga} + 1023.3$ |
| | | | $G^{Ga_2Gd}_{Gd:Gd} = +GHSER_{Gd}+34004.3$ |
| | | | ${}^0L^{Ga_2Gd}_{Ga,Gd:Ga} = {}^0L^{Ga_2Gd}_{Ga,Gd:Gd} = 0.0$ |
| | | | ${}^0L^{Ga_2Gd}_{Ga:Ga,Gd} = -16024.7 - 3.4105T$ |
| | | | ${}^0L^{Ga_2Gd}_{Gd:Ga,Gd} = +52110.7$ |
| | $Ga_2Gd_3$ | $(Ga)_{0.4}(Gd)_{0.6}$ | $G^{Ga_2Gd_3}_{Ga:Gd} = +0.4GHSER_{Ga} + 0.6GHSER_{Gd} - 56006.0+9.5292T$ |
| | $Ga_3Gd_5$ | $(Ga)_3(Gd)_5$ | $G^{Ga_3Gd_5}_{Ga:Gd} = +0.375GHSER_{Ga} + 0.625GHSER_{Gd} - 552493.1+8.5320T$ |
| | $Ga_6Gd$ | $(Ga)_{0.857143}$ $(Gd)_{0.142857}$ | $G^{Ga_6Gd}_{Ga:Gd} = +0.857143GHSER_{Ga} + 0.142857GHSER_{Gd} - 31929.9+2.3540T$ |
| | GaGd | $(Ga)_{0.5}(Gd)_{0.5}$ | $G^{GaGd}_{Ga:Gd} = +0.5GHSER_{Ga} + 0.5GHSER_{Gd} - 62480.0+10.0116T$ |

33.3 at.% Gd [25, 30]. The metastable phases $Ga_{21}Gd_{79}$ and $Ga_{25}Gd_{75}$ were found from the crystallization of amorphous alloys obtained with melt spinning by Buschow et al. [31]. The magnetic properties and thermal stability of these phases were investigated [31]. Several magnetic property measurements were carried out for all Ga–Gd compounds, mostly in the 10–300 K temperature range [18, 32–39]. As the reported magnetic transitions are rather under at 300 K, these experimental data [18, 32–39] are not utilized in the present modeling.

The thermodynamic properties of Ga–Gd alloys were determined by several researchers [16, 40–42]. The enthalpies of formation of $Ga_2Gd$ were determined at 1373 K [16] and at 823 K [41]. The Gibbs energy, enthalpy, and entropy of formation of Ga and Gd were calculated and partial molar thermodynamic quantities were derived by Shkolnikova et al. [40], which was cited from Ref. [17]. But the original literature [40] was not found with every possible means. The enthalpy, entropy, and Gibbs energy of formation of $Ga_2Gd$ from liquid Ga and solid Gd at 550 °C were as: − 261.5 ± 5.9 kJ/mol, − 88.3 ± 7.1 J/mol, and − 189.0 ± 0.4 kJ/mol [42], respectively, which was cited from Ref. [17]. These results [42] were too negative and not considered in the present optimization.

## Results and discussions

### The Ce–Ga system

A thermodynamic description of the Ce–Ga system obtained in the present work is shown in Table 1. During the process of the optimization of the Ce–Ga system, we set a more weight on the results reported by Massalski [4]. The phase diagram of the Ce–Ga system calculated by means of the present thermodynamic parameters is presented in Fig. 2. The calculated phase diagram of the Ce–Ga system is compared with the experimental data by Dayan et al. [7], Kimmel et al. [8], Yatsenko et al. [6, 9], and the assessed results by Massalski [4] and Idbenali et al. [3]. The present calculated results are more close to the experimental data [6–9] and the assessed results [4], compared with the assessed liquid by Idbenali et al. [3]. The assessed results of the composition from 50 to 90 at.% Ga are fitting well with the experimental data reported by Dayan et al. [7], Kimmel et al. [8], and Yatsenko et al. [6, 9] in the present work, as shown in Fig. 2. A small discrepancy is observed for the composition from 25 to 50 at.% Ga of the liquidus line, which is a little higher than the experimental data [6]. The Ga-rich side of the liquidus is a little underestimated compared with the assessed results [4] and the experimental data [6–9]. The reasonable solid solution of the $CeGa_2$ with homogeneity ranging from about 66.67 to 77.94 at.% Ga in the present work is consistent with the results reported by Massalski [4] and Kimmel and Dayan [8].

The invariant equilibria of the Ce–Ga system are listed in Table 2. There are nine invariant reactions in the Ce–Ga system. The invariant reaction temperatures show a scattering of up to 52 K in the reaction liq. → $Ce_3Ga$ + fcc(Ce), compared with the experimental data [3] and the assessed results [4]. eight of the nine experimental invariant reaction temperatures in the Ce–Ga system are well reproduced, as shown in Table 2. In view of the estimated experimental errors, the Ce–Ga system are well reproduced, close to the phase diagram reported by Massalski [4] and nearly like to the phase diagram assessed by Idbenali et al. [3].

Figure 3 presents the calculated enthalpies of the formation at 300 K in the Ce–Ga system with the experimental data reported by Yatsenko et al. [9], Vnuchkova et al. [10], Yamshchikov et al. [11], Chkhikvadze and Kereselidze [13], Boyer et al. [14], Babu et al. [15], Meschel and Kleppa [16], and the assessed results by Idbenali et al. [3], and the values calculated from Materials Project database [43]. The reference states for the elements are orth. for Ga and fcc for Ce. As shown in Fig. 3, the present assessed results are more negative than the experimental data [10, 11, 13, 14, 16, 43] and the assessed results by Idbenali et al. [3], while they are close to the experimental data reported by Yatsenko et al. [9] and Babu et al. [15]. All the calculated values by the Materials Project database [43] are less negative than the experimental data [3, 9–11, 13–15].

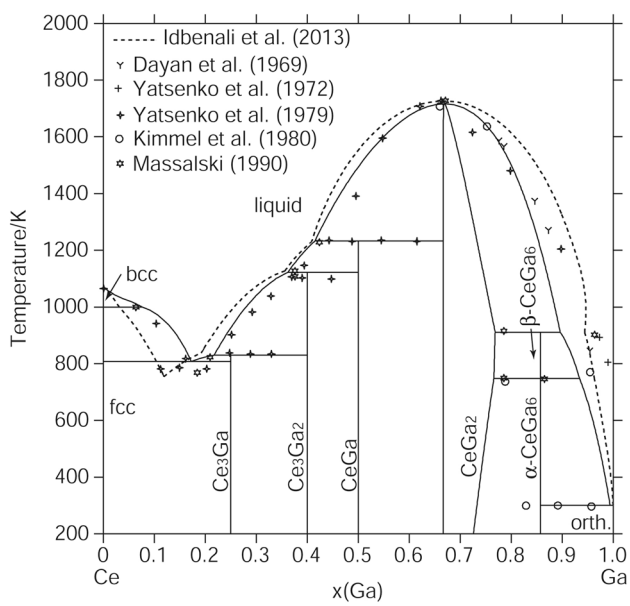


**Fig. 2** Calculated Ce–Ga phase diagram by the thermodynamic description with the experimental data by Dayan et al. [7], Kimmel et al. [8], Yatsenko et al. [6, 9], and the assessed results by Massalski [4] and Idbenali et al. [3]

**Table 2** Invariant reactions of the Ce–Ga system

| Reaction | *T*/K | *x*/Ga | | | References |
|---|---|---|---|---|---|
| liq. → $Ce_3Ga$ + fcc (Ce) | 807 | 0.1726 | 0.2500 | 0.0 | This work |
| | 768 | 0.1800 | – | – | Massalski [4] |
| | 755 | 0.1190 | – | – | Idbenali et al. [3] |
| | 773 | 0.6667 | – | – | Yatsenko et al. [6] |
| liq. + bcc (Ce) → fcc(Ce) | 1000 | 0.0767 | – | – | This work |
| | 999 | 0.0580 | – | – | Massalski [4] |
| | 1000 | 0.0310 | – | – | Idbenali et al. [3] |
| liq + $Ce_3Ga_2$ → $Ce_3Ga$ | 830 | 0.2172 | – | – | This work |
| | 830 | 0.2070 | – | – | Massalski [4] |
| | 837 | 0.1922 | – | – | Idbenali et al. [3] |
| liq + CeGa → $Ce_3Ga_2$ | 1123 | 0.3623 | 0.5000 | 0.6667 | This work |
| | 1123 | 0.3700 | – | – | Massalski [4] |
| | 1127 | 0.3564 | – | – | Idbenali et al. [3] |
| liq + $CeGa_2$ → CeGa | 1233 | 0.4150 | 0.6667 | 0.5000 | This work |
| | 1233 | 0.4210 | – | – | Massalski [4] |
| | 1231 | 0.4086 | – | – | Idbenali et al. [3] |
| liq → $CeGa_2$ | 1717 | 0.6667 | – | – | This work |
| | 1733 | 0.6667 | – | – | Yatsenko et al. [6] |
| | 1727 | 0.6667 | – | – | Massalski [4] |
| | 1727 | 0.6667 | – | | Idbenali et al. [3] |
| | 1717 | – | – | – | Dayan et al. [7] |
| liq + $CeGa_2$ → β-$CeGa_6$ | 910 | 0.8958 | 0.7794 | 0.8571 | This work |
| | 910 | 0.9560 | – | – | Massalski [4] |
| | 911 | 0.9438 | – | – | Idbenali et al. [3] |
| liq + $CeGa_2$ → α-$CeGa_6$ | 748 | 0.9337 | 0.7778 | 0.8571 | This work |
| | 750 | 0.9770 | – | – | Massalski [4] |
| | 750 | 0.9523 | – | – | Idbenali et al. [3] |
| liq → α-$CeGa_6$ + orth.(Ga) | 300 | 0.9937 | 0.8571 | 1.0000 | This work |
| | 302 | 1.0000 | – | – | Massalski [4] |
| | 303 | 0.9995 | – | – | Idbenali et al. [3] |

Figure 4a, b presents the calculated enthalpies of the mixing of the liquid in the Ce–Ga system with the assessed results by Idbenali et al. [3] and the experimental data determined by Babu et al. [15] at 876 K and 1125 K, respectively. The reference states for the elements are liquid for Ce and Ga. A satisfactory agreement is reproduced at 876 K and 1125 K in this work and comparison with the experimental data [15], and close to the results assessed by Idbenali et al. [3], as shown in Fig. 4a, b. The optimized partial mixing enthalpies are more negative in the present work, and we did not obtain a better agreement with the experimental data [9–14], but a bit consistent with the results assessed by Idbenali et al. [3]. Comparison with the assessed Ce–Ga phase diagram, some improvements are obtained based on the experimental data in the present work, the liquidus line is more close to the most experimental points, and the homogeneity range of the compound $CeGa_2$ seems more reasonable at full temperature range.

## The Ga–Gd system

A thermodynamic description of the Ga–Gd system obtained in the present work is shown in Table 1. During the process of the optimization of the Ga–Gd system, we set a more weight on the results reported by Palenzona and Clrafici [17].

The invariant equilibria of the Ga–Gd system are listed in Table 3. As shown in Table 3, satisfactory agreement is obtained between the calculations and experiments by Palenzona and Clrafici [17]. There are eight invariant reactions in the Ga–Gd system, and the reaction temperatures of the five invariant reactions are the same as the results reported by Palenzona and Clrafici [17]. The temperature of the reaction Liq. → hcp + $Ga_3Gd_5$ is 1114 K, which is a little lower than the results [9, 17], and nearly close to the result [18]. In view of the estimated experimental errors, the invariant reactions of the Ga–Gd system are well reproduced, similarly close to the reported phase diagram [9, 17, 22].

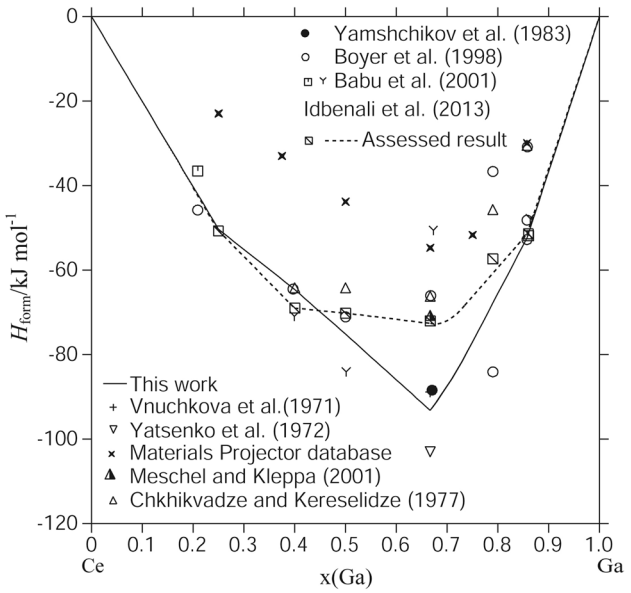


**Fig. 3** Calculated enthalpies of the formation at 300 K in the Ce–Ga system with the experimental data by Yatsenko et al. [9], Vnuchkova et al. [10], Yamshchikov et al. [11], Chkhikvadze and Kereselidze [13], Boyer et al. [14], Babu et al. [15], Meschel and Kleppa [16], and the assessed results by Idbenali et al. [3], and the value calculated from Materials Projector database [43]. The reference states for the elements are orth. for Ga and fcc for Ce

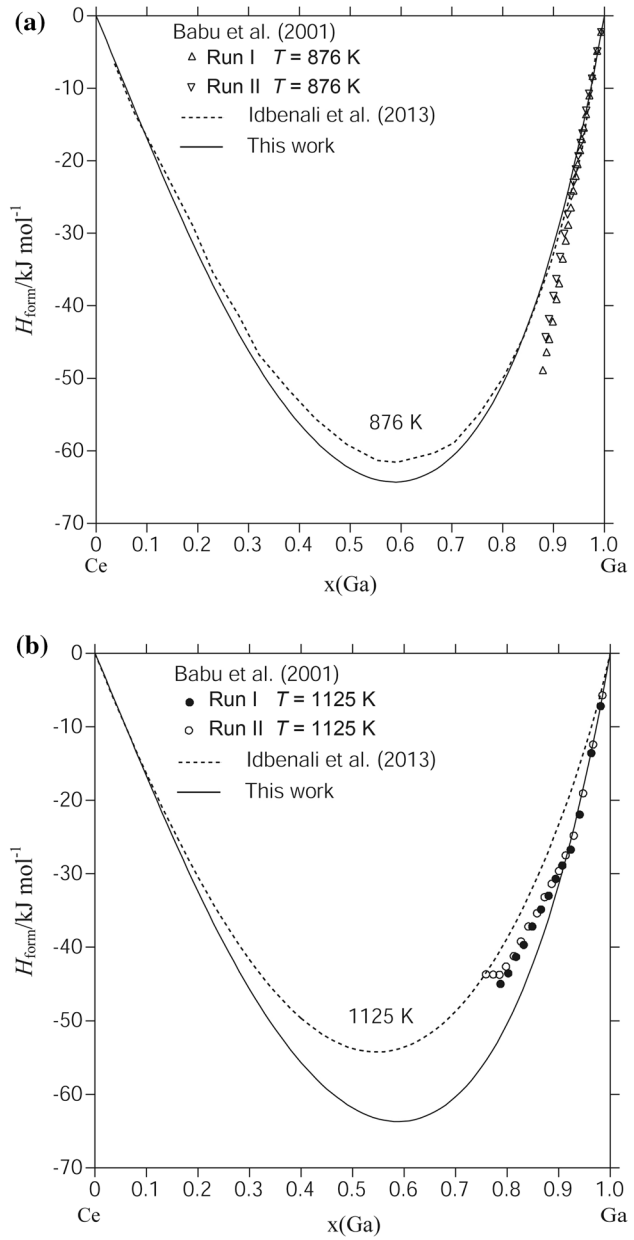


**Fig. 4** **a** Calculated mixing enthalpies of liquid at 876 K in the Ce–Ga system and comparison with the experimental data determined by Babu et al. [15] and the assessed result by Idbenali et al. [3]. The reference states for the elements are liquid for Ce and Ga. **b** Calculated mixing enthalpies of liquid at 1125 K in the Ce–Ga system and comparison with the experimental data determined by Babu et al. [15] and the assessed result by Idbenali et al. [3]. The reference states for the elements are liquid for Ce and Ga

The phase diagram of the Ga–Gd system calculated by means of the present thermodynamic parameters is presented in Fig. 5, which is compared with the experimental data by Yatsenko et al. [6, 9, 19], Pelleg and Zevin [20], and Buschow and Hoogenhof [18]. A small discrepancy is observed for the composition from 70 to 85 at.% Gd of the liquidus, which is a little higher than the experimental data determined by Yatsenko et al. [6, 19] and Buschow and Hoogenhof [18]. The reasonable solid solution of the $Ga_2Gd$ with a homogeneity domain ranging from about 19.64 to 33.33 at.% Gd in the present work is consistent with the results reported by Palenzona and Clrafici [17]. The calculated phase diagram of the Ga–Gd system is nearly identical to the Ga–Gd phase diagram constructed by Palenzona and Clrafici [17]. The enthalpy, entropy and Gibbs energy of formation of $Ga_2Gd$ from liquid Ga and solid Gd at 550 °C were too negative: − 261.5 ± 5.9 kJ/mol, − 88.3 ± 7.1 J/mol, and − 189.0 ± 0.4 kJ/mol [42], which are very largely different from the calculated values from the Materials Project database [43]. So the thermodynamic experimental data of $Ga_2Gd$ is not used during the process of the optimization in the present work. Figure 6 is the enthalpies of formation from the present CALPHAD modeling at 300 K and 1373 K by comparison with the experimental data by Bayanov and Ganchenko [41] at 873 K, Meschel and Kleppa [16] at 1373 K, and the enthalpies of formation at 0 K in the Materials Project database [43]. The reference states at 300 K are orth. for Ga and hcp_A3 for Gd, at 1373 K are liquid for Ga and hcp_A3 for Gd. In the present work, the values of the Materials Project database [43] are set more weight during the process of optimization. The trend of the assessed results is consistent with the value of the Materials Project database [43].

**Table 3** Invariant reactions of the Ga–Gd system

| Reaction | *T*/K | *x*/Gd | | | References |
|---|---|---|---|---|---|
| liq. + bcc → hcp(Gd) | 1535 | 0.9795 | 1.0000 | 1.0000 | This work |
| | 1508 | – | 1.0000 | 1.0000 | Palenzona and Clrafici [17] |
| liq. + $Ga_2Gd$ → GaGd | 1468 | 0.5225 | 0.3333 | 0.5000 | This work |
| | 1468 | 0.5500 | 0.3333 | 0.5000 | Palenzona and Clrafici [17] |
| | 1473 | – | – | – | Yatsenko et al. [9] |
| | 1463 | – | – | – | Pelleg and Zevin [22] |
| | 1463 | – | – | – | Buschow and Hoogenhof [18] |
| liq. → $Ga_2Gd$ | 1673 | 0.3333 | – | – | This work |
| | 1673 | 0.3333 | – | – | Yatsenko et al. [9] |
| | 1673 | 0.3333 | – | – | Pelleg and Zevin [22] |
| | 1673 | 0.3333 | – | – | Buschow and Hoogenhof [18] |
| liq. + GaGd → $Ga_2Gd_3$ | 1383 | 0.6340 | 0.5000 | 0.6000 | This work |
| | 1383 | 0.6150 | 0.5000 | 0.6000 | Palenzona and Clrafici [17] |
| | 1393 | – | – | – | Yatsenko et al. [9] |
| | 1373 | – | – | – | Buschow and Hoogenhof [18] |
| liq. + $Ga_2Gd_3$ → $Ga_3Gd_5$ | 1283 | 0.7465 | 0.6000 | 0.6250 | This work |
| | 1283 | – | – | – | Predel [44] |
| | 1363 | 0.6400 | 0.6000 | 0.6250 | Palenzona and Clrafici [17] |
| | 1373 | – | – | – | Yatsenko et al. [9] |
| | 1352 | – | – | – | Buschow and Hoogenhof [18] |
| liq. → hcp + $Ga_3Gd_5$ | 1114 | 0.8479 | 0.6250 | 1.0000 | This work |
| | 1123 | – | – | – | Yatsenko et al. [9] |
| | 1123 | 0.8000 | 0.625 | 1.0000 | Palenzona and Clrafici [17] |
| | 1118 | – | – | – | Buschow and Hoogenhof [18] |
| liq. + $Ga_2Gd$ → $Ga_6Gd$ | 679 | 0.0006 | 0.1964 | 0.1429 | This work |
| | 679 | – | – | – | Pelleg and Zevin [22] |
| | 679 | 0.4000 | 0.2200 | 0.1430 | Palenzona and Clrafici [17] |
| liq. + $Ga_6Gd$ → orth. | 303 | 0.0 | 0.1429 | 0.0 | This work |
| | 303 | – | – | – | Yatsenko et al. [9] |
| | 302 | – | – | – | Pelleg and Zevin [22] |
| | 302 | < 0.1 | 0.1430 | 0.0 | Palenzona and Clrafici [17] |

## Conclusions

The phase relations and the thermodynamic modeling of the Ce–Ga and Ga–Gd systems are critically evaluated from the experimental data available in the literature. A set of consistent thermodynamic parameters of the Ce–Ga and Ga–Gd systems is derived, respectively. In the Ce–Ga system, there are nine invariant reactions, and five stoichiometric intermetallic compounds and the compound $CeGa_2$ with a homogeneity domain ranging from about 66.67 to 77.94 at.% Ga. In the Ga–Gd system, there are eight invariant reactions, and four stoichiometric intermetallic compounds and the compound $Ga_2Gd$ with a homogeneity domain ranging from about 19.64 to 33.33 at.% Gd. With the thermodynamic modeling available, one can now make various calculations of practical applications, especially for the design of the magnetic materials or in the optimization of high-order systems.

## Methods

### Thermodynamic models

The Gibbs energy function for the elements, Ce, Ga, and Gd in the phase, liquid, orthorhombic (orth.), fcc, bcc, and hcp is taken from the SGTE (Scientific Group Thermodata Europe) pure elements database compiled by Dinsdale [45] and described as the same as our previous paper [46]. The Gibbs energies of the five solution phases, liquid, orth., bcc, fcc, and hcp are described as the same as our previous published paper [46], as shown in Table 1.

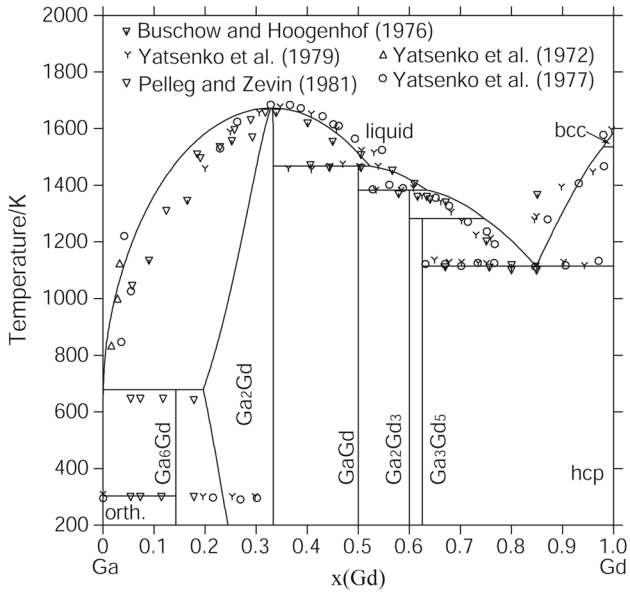


**Fig. 5** Calculated Ga–Gd phase diagram by the thermodynamic description with the experimental data by Yatsenko et al. [6, 9, 19], Pelleg and Zevin [20], and Buschow and Hoogenhof [18]

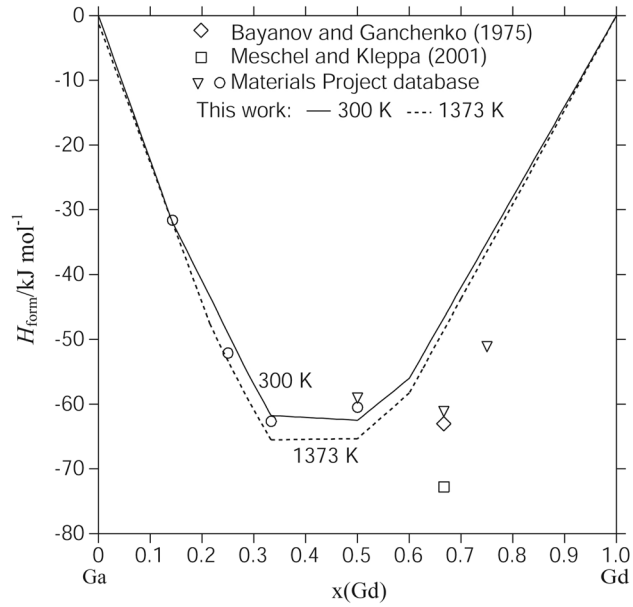


**Fig. 6** Enthalpies of formation from the present CALPHAD modeling at 300 K (solid line) and 1373 K (dotted line) by comparison with the experimental data by Bayanov and Ganchenko [41] at 873 K, Meschel and Kleppa [16] at 1373 K, and the enthalpies of formation at 0 K in the Materials Project database [43]. The reference states at 300 K are orth. for Ga and hcp_A3 for Gd, at 1373 K are liquid for Ga and hcp_A3 for Gd

In the Ce–Ga and Ga–Gd systems, the nine intermetallic compounds, CeGa, $Ce_3Ga_2$, $Ce_3Ga$, α-$CeGa_6$, β-$CeGa_6$, GaGd, $Ga_6Gd$, $Ga_2Gd_3$, and $Ga_3Gd_5$ are treated as stoichiometric compounds. The two intermetallic compounds $CeGa_2$ and $Ga_2Gd$ with a homogeneity range are treated as non-stoichiometric intermetallic compounds. In the present work, all the compounds in the two systems are used a two-sublattice model, which are similar to our previous published paper [46], as shown in Table 1.

## Optimization

Most of the above experimental data [3, 4, 6–11, 13–15, 43] in the Ce–Ga system are selected for the evaluation of the thermodynamic modeling. In the present work, the phase relations and transformation temperatures of the Ce–Ga system reported by Massalski [4] are mainly considered, the same as the treatment by Idbenali et al. [3]. The experimental data by Yatsenko et al. [6, 9, 19], Pelleg and Zevin [20], and Buschow and Hoogenhof [18] in the Ga–Gd phase diagram are selected for the evaluation of the thermodynamic modeling. The assessment of the phase diagram of the Ce–Ga and Ga–Gd systems are carried out by means of the optimization module PARROT of the thermodynamic software Thermo-Calc [47], which can deal with various kinds of experimental information.

For the liquid, bcc, fcc, hcp, and orth. solutions in the Ce–Ga and Ga–Gd systems, the interaction parameters ${}^0L^{liq}$, ${}^1L^{liq}$, ${}^2L^{liq}$ or ${}^3L^{liq}$ in Eq. (6) can be reliably obtained from the experimental data [4, 6–9] in the Ce–Ga system, and the experimental data [6, 9, 18–20] in the Ga–Gd system.

In the Ce–Ga and Ga–Gd systems, the intermetallic compounds, CeGa, $Ce_3Ga_2$, $Ce_3Ga$, α-$CeGa_6$, β-$CeGa_6$, GaGd, $Ga_6Gd$, $Ga_2Gd_3$, and $Ga_3Gd_5$ are treated as stoichiometric compounds with a two-sublattice model.

The compound $CeGa_2$ has a homogeneity ranging from about 66.67 to 78 at.% Ga reported by Massalski [4] and Kimmel and Dayan [8]. The solid solution of the $Ga_2Gd$ has a homogeneity ranging from about 22 to 33.33 at.% Gd reported by Palenzona and Clrafici [17]. So in the present work, the intermetallic compounds $CeGa_2$ and $Ga_2Gd$ are treated using a two-sublattice model with the formula $(Ce,Ga)_{0.333333}(Ce,Ga)_{0.666667}$, and $(Ga,Gd)_{0.666667}(Ga,Gd)_{0.333333}$, respectively.

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